

Semester: I /II					
Course Title: Engineering Explorations (Theory and Lab) Category: Multidisciplinary					
Course Code	:	CS1841	CIE	:	70 Marks
Credits	:	3	SEE	:	30 Marks
L:T:P Hours		2:0:2			
Total Hours	:	30L+30P	SEE Duration	:	2:00 Hours

Course Overview The course "Engineering Explorations" is designed to provide a comprehensive introduction to engineering visualization, manufacturing techniques, and emerging technologies in engineering. The syllabus begins with the fundamentals of engineering drawing, including BIS conventions, freehand sketching, and an overview of Computer-Aided Drafting (CAD) software. The course covers orthographic projections of points, lines, planes, and solids using first-angle projection methods to develop spatial reasoning. Isometric projection techniques are taught to represent three-dimensional objects accurately. An introduction to Additive Manufacturing (AM) familiarizes students with 3D printing processes, materials, and post-processing methods. Practical sessions include modeling and printing geometric shapes like triangles, pentagons, and hexagons. Students also engage in interdisciplinary 2D drafting and 3D printing projects.

Module -I	6 Hrs
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Introduction to Engineering Visualisation

Importance of Engineering drawing, BIS Conventions of Engineering Drawing, Freehand sketching of engineering drawing, Scales (Only introduction, Not for CIE & SEE). Introduction to Computer Aided Drafting software, Coordinate system and reference planes HP, VP, RPP & LPP of 2D/3D environment. Selection of drawing sheet size and scale.

Orthographic Projections of Points and Lines

Introduction to Orthographic projections: Orthographic projections of points in 1st and 3rd quadrants. Orthographic projections of lines (Placed in First quadrant only).

Orthographic Projections of Planes

Orthographic projections of planes viz triangle, square, rectangle, pentagon, hexagon, and circular laminae (Placed in First quadrant only using change of position method).

Module – II**6 Hrs****Orthographic Projections of Solids:**

Orthographic projection of right regular solids (Solids Resting on HP only): Prisms & Pyramids (triangle, square, rectangle, pentagon, hexagon, cube).

Module –III**6 Hrs**

Isometric Projections: Isometric scale, Isometric projection of hexahedron (cube), right regular prisms, pyramids, cylinders, cones and spheres. Isometric projection of a combination of two simple solids.

Project: Application of 2D drawing to interdisciplinary applications: Create a software drawing for real-world objects.

Module –IV**6 Hrs**

Introduction to Additive Manufacturing Process: Generic Additive Manufacturing (AM) process, the benefits of Additive Manufacturing, the steps in additive manufacturing & importance. Techniques in Additive Manufacturing, Materials and 3D Printing Workflow, Post-processing, Quality Checking & Applications. Project: 3D Printing of Geometric Shapes by Reverse Engineering.	
Module –V	6 Hrs
CAED for Additive Manufacturing Introduction to CAED modeling in 3D printing. Introduction to CAD environment, Sketching, Modeling and Editing features, Different file formats, Export/Import geometries, Part orientation, Layer slicing, Process path selection, Printing. Project: 3D Printing of Geometric Shapes like Triangle, Pentagon, and Hexagon	

Course Outcomes: After completing the course, the students will be able to: -

CO 1	Apply BIS conventions, freehand sketching, orthographic projections, and computer-aided drafting concepts to visualize and represent engineering components in 2D and 3D.
CO 2	Construct orthographic projections of points, lines, planes, and solids using first-angle projection principles, demonstrating a clear understanding of spatial visualization.
CO 3	Evaluate isometric projections of various solids and their combinations using isometric scale and projection techniques to represent real-world 3D objects.
CO 4	Demonstrate the use of CAED tools in additive manufacturing processes by creating, modeling, and 3D printing simple geometric shapes using proper slicing, orientation, and post-processing methods.

Textbook Books	
1	Engineering Drawing by N.D. Bhatt. 53rd Edition. Charotar Publishing House. ISBN-10: 9385039270. ISBN-13: 978-9385039275.
2	Additive Manufacturing: Materials, Processes, Quantifications, and Applications by Kun Zhou. 1st Edition. CRC Press. ISBN-10: 1138333448. ISBN-13: 978 1138333444.
3	Introduction to Robotics: Mechanics and Control by John J. Craig. 4th Edition. Pearson Education. ISBN-10: 1292040041. ISBN-13: 978-1292040042.



Reference Books	
1.	Engineering Graphics with AutoCAD by James D. Bethune. 1st Edition. Pearson Education. ISBN-10: 0134251648. ISBN-13: 978-0134251649.
2.	Additive Manufacturing Technologies: 3D Printing, Rapid Prototyping, and Direct Digital Manufacturing by Ian Gibson, David W. Rosen, and Brent Stucker. 2nd Edition. Springer. ISBN-10: 1493921134. ISBN-13: 978-1493921133.
3.	Introduction to Mechatronics and Measurement Systems by David G. Alciatore and Michael B. Hstand. 5th Edition. McGraw-Hill Education. ISBN-10: 1259675479. ISBN-13: 978-1259675472.

LABORATORY COMPONENT

1.	Orthographic Projections of Solids
2.	Isometric Projection of Solids
3.	Application of 2D drawing to Interdisciplinary Application
4.	Learn to create 3D models using solid-edge software and prepare them for printing using slicing software.
5.	Create the Engineering model by using reverse engineering Applications and printing

	the same with 3D printers.
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